

US Food Access & Senior Hunger Gap

An Interactive Power BI Dashboard

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Where This Idea Came From

I grew up in India. Poverty and food insecurity are visible realities there -not statistics, but daily life for millions. So, when I heard a story at a church service about a food initiative helping people in need, it resonated with me immediately. The initiative was doing something meaningful: providing free food to those who could not afford it.

But there was a gap the initiative could not close. People who were homebound-bedridden seniors, individuals with disabilities, those too ill to travel - could not reach the food. The church could provide it. The community wanted to give it. But there was no way to get it to the people who needed it most.

"The food existed. The willingness existed. The gap was access."

I carried that observation with me when I moved to Denton, Texas to pursue my Master's degree. And here is where the real question hit me:

India is a developing country. Everyone understands that poverty and food insecurity exist there. But I was now living in Texas, in the United States, one of the wealthiest states in the world's wealthiest nation. Denton County alone has a population of nearly one million people and is one of the fastest-growing suburban counties in America.

"Does the same problem exist here? In my own city?"

I did not want to assume. I did not want to guess. I wanted data. So I went looking for it.

The Data Source

After researching publicly available government datasets, I identified the USDA Economic Research Service Food Environment Atlas - a comprehensive, nationally recognized dataset covering food access conditions across every county in the United States. This was the right source because:

- It is published by a federal government agency -not a private organization or advocacy group
- It covers all 3,000+ US counties, enabling national-to-local analysis
- It includes variables specifically measuring senior food access at the county level
- It is freely available with no registration required
- It is updated regularly -the version I used was released July 2025

I downloaded the CSV files directly from ers.usda.gov. No third-party aggregators, no web scraping, no synthetic data. Just raw government numbers.

The dataset contains over 300 variables. I identified 6 that were directly relevant to my question.

Variable	What It Measures	Level
FOODINSEC_21_23	% of households facing food insecurity, 2021-23	State
FOODINSEC_18_20	% of households facing food insecurity, 2018-20	State
PCT_LACCESS_SENIORS19	% of seniors with low food store access, 2019	County
LACCESS_SENIORS19	Count of seniors with low food store access, 2019	County
DEEPPPOVRATE21	Deep poverty rate, 2021	County
PCT_SNAP22	% of population receiving SNAP benefits, 2022	County

What the National Data Shows

Before narrowing to Denton, I analyzed the data at the national level across all 3,000+ counties. Three patterns stood out immediately:

01

Rural counties face near-total access collapse

Counties like Loving County, TX show nearly 100% of seniors without food store access. This is not a resource problem -it is a geographic isolation problem. Sparse populations and long distances between stores create conditions where food access is structurally impossible without home delivery programs.

02

Deep poverty and food access gaps are strongly correlated

Across all US counties, the highest deep poverty rates consistently align with the worst senior food access gaps. Targeted investment in food delivery infrastructure in high-poverty counties would have disproportionately high impact.

03

Food insecurity is getting worse nationally, not better

Comparing 2018-20 to 2021-23, food insecurity rates increased across the country. The pandemic-era assumption that conditions would improve post-2020 is not reflected in the data.

What I Found in Denton, TX

When I filtered the dashboard to Texas and then specifically to Denton County -my home -the numbers were more troubling than I expected.

17,464 Seniors without food access in Denton	16.9% Food insecurity rate -up 27% in 3 years	37.93% Denton seniors with low food access	11.5% On SNAP -vs 16.9% food insecure
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Finding 1 - Food insecurity is rising in one of America's wealthiest suburbs

Denton County is not a poor county. It is a high-growth suburban area with nearly 979,000 residents, major universities, and a strong job market. Yet food insecurity rose from 13.3% in 2018-20 to 16.9% in

2021-23 -a 27% increase in just three years. Economic growth in Denton is clearly not reaching everyone. The gap between prosperity and food security is widening, not narrowing.

Finding 2 - Nearly 4 in 10 Denton seniors cannot reach a food store

37.93% of seniors in Denton County have low food store access -that is 17,464 real individuals. For a suburban county with extensive road infrastructure and retail development, this number is striking. The data suggests this is not a supply problem. Food stores exist in Denton. The problem is that seniors cannot physically get to them -whether due to mobility limitations, lack of transportation, or being homebound entirely.

This is exactly the gap I heard about in India. The food exists. The people need it. The barrier is access -and for homebound seniors, that barrier is insurmountable without a delivery program.

Finding 3 - Most food-insecure Denton residents are not in extreme poverty

Denton's deep poverty rate is only 3.58% -yet food insecurity stands at 16.9%. This gap reveals something important: the majority of food-insecure residents in Denton are not the stereotypical image of poverty. They are working households, fixed-income seniors, and middle-income families who earn too much to qualify for assistance programs but too little to consistently afford food.

Finding 4 -There is a measurable assistance gap

16.9% of Denton households face food insecurity. Only 11.5% receive SNAP benefits. That gap - approximately 5 percentage points -represents households that are food insecure but are not being captured by any federal assistance program. They are invisible to the system designed to help them.

The Conclusion

"The same problem I heard about in India exists in Denton, Texas. The scale is different. The wealth of the surrounding environment is different. But the gap - between people who need food and people who can actually access it -is the same."

16.78 million seniors across the United States lack food store access. In Denton alone, that is 17,464 people. These are not abstractions. These are neighbours, grandparents, and community members who cannot get to a store -and in many cases, have no program that will bring food to them.

Recommendations for Further Analysis

This dashboard identifies where the gap exists and how large it is. But data alone does not feed anyone. Closing the gap identified here would require a second phase of analysis that this project intentionally scopes out — but makes possible.

1. Map existing program coverage against identified need zones

Programs like Meals on Wheels and SPAN in Denton County deliver meals to homebound seniors Monday through Friday. The next analytical question is: which ZIP codes have the highest senior food access gaps but the lowest program coverage? Overlaying delivery program service areas against PCT_LACCESS_SENIORS19 data by ZIP code would surface the most underserved zones immediately.

2. Quantify the weekend delivery gap

Current programs in Denton operate only on weekdays. A senior who receives meals Monday through Friday still faces a 2-day gap every week. Quantifying how many homebound seniors fall into this weekend gap — and what it would cost to close it — is a measurable, actionable analysis.

3. Identify the hidden food-insecure population

The gap between food insecurity rate (16.9%) and SNAP participation (11.5%) in Denton points to a population that needs help but is not in any system. Further analysis using Census income bracket data could help identify which household types — working families, fixed-income seniors, single-parent households — make up this invisible population.

4. Model last-mile delivery feasibility for rural counties

Counties like Loving County, TX with nearly 100% senior food access gaps are not failures of willpower — they are failures of infrastructure. A route optimization analysis using county geography and population density data could determine whether community hub models or mobile delivery units are more feasible for extremely rural areas.

The Bottom Line

This dashboard answers the question: "How bad is the problem?" The answer is — worse than most people living in a growing, prosperous suburb like Denton, TX would expect.

The next question is: "What would it take to fix it?" That question deserves its own data project — and this dashboard is the foundation for it.

The Dashboard

I built this interactive Power BI dashboard to make these findings explorable -not just for Denton, but for any county in the United States. Using State and County slicers, any user can filter all six visuals simultaneously to investigate food insecurity conditions in their own community.

Visual	What It Shows
4 KPI Cards	Snapshot of key metrics for selected state/county -food insecurity %, senior access %, deep poverty %, SNAP participation %
Senior Food Access Bar Chart	Counties ranked by % of seniors with low food access -surfaces the worst-affected areas instantly
Deep Poverty Bar Chart	Counties ranked by deep poverty rate -identifies most economically vulnerable communities
Food Insecurity Trend	2018-20 vs 2021-23 comparison -shows whether conditions are improving or worsening
Metric Summary Table	All 6 metrics for the selected filter -complete picture in one view
Seniors Without Food Access	Live count of affected seniors -translates percentages into real human numbers

About Me

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Data Analyst with 3+ years of experience managing and validating data to ensure accuracy and integrity. Skilled in daily data validation, reporting, and dashboard development using Power BI and Excel, with SQL and Python for data transformation. Proven ability to collaborate with operations teams to capture critical metrics and communicate actionable insights to stakeholders.

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Available at: <https://www.ers.usda.gov/data-products/food-environment-atlas/>